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Speed Control of a Three-Phase Brushless Direct Current Motor: Developing Mathematical Modelling



Abstract: In recent years, variable speed drive systems have gained universal adoption across both small-scale and large-scale applications, such as the automotive industry and household appliances. With a growing stress on energy efficiency and environmentally friendly technologies, Brushless DC Motors (BLDCMs) have emerged as an important alternative due to their superior performance characteristics. These include high efficiency, improved low-speed torque, high power density, reduced maintenance, and lower noise than conventional motors like induction and reluctance motors. This paper presents the mathematical modelling of a three-phase BLDC motor, and a PID controller is implemented to achieve closed-loop speed regulation under varying load conditions, demonstrating stable and reliable motor control.

Keywords: *Brushless DC motor, Back EMF, Hall sensors, PI controller, electromagnetic torque.*

Introduction

Electric motors consume nearly half of the electrical energy used worldwide. In most appliances, the commonly used motors are either single-phase induction motors or brushed DC motors, both of which suffer from low efficiency and require high maintenance. Single-phase induction motors are less efficient due to ohmic losses in the rotor. However, these losses can be significantly reduced in Brushless DC (BLDC) motors, thanks to the absence of brushes and mechanical commutation [1].

BLDC motors are a type of synchronous motor, meaning the rotor's magnetic field and the stator's magnetic field rotate at the same frequency. They are available in single-phase, two-phase, and three-phase configurations, with the three-phase type being the most widely used [2]. A BLDC motor typically features a permanent magnet rotor and wire-wound stator poles. Since it lacks brushes and a mechanical commutator, it requires an electronic inverter for commutation, along with a rotor position sensor. Hall-effect sensors are commonly employed to detect the rotor position.

The inverter uses transistors for low-power drives and thyristors for high-power drives. Hall-effect sensors are typically mounted on the motor shaft to detect rotor position [3]. Most BLDC motors incorporate three Hall-effect sensors [4]. As the rotor's magnetic poles pass near the sensors, they generate high or low signals, indicating the presence of either a north or south pole.

The construction of conventional BLDC motors is widely used in industries such as robotics, aerospace, automotive, medical devices, and industrial automation equipment

They offer several advantages over brushed DC motors and induction motors, including:

1. Reduced maintenance requirements due to the absence of brushes,
2. Lower electrical noise generation,
3. Low rotor inertia, leading to improved dynamic response,
4. Superior speed–torque characteristics,
5. Extended operational life, and

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6. High efficiency

Mathematical Model of BLDC Motor

In the BLDC motor shown in Fig. 1 three stator windings are connected in a star (Y) configuration. The motor is powered by a three-phase voltage source, which can supply not only sinusoidal voltages but also square waves or other voltage waveforms.

The modeling of the BLDC motor is based on the following assumptions:

1. The motor operates without magnetic saturation and under-rated current conditions.
2. The resistances of all three stator windings are identical.
3. Self-inductance and mutual inductance are constant.
4. Iron and stray losses are negligible.
5. The three-phase system is balanced.
6. The air gap is uniform.
7. Hysteresis and eddy current losses are neglected.
8. Semiconductor switches are considered ideal.

The armature winding model of the BLDC motor can be expressed by the following equations. Thus, the complete dynamic model of the BLDC motor can be represented using these equations.

$$V_a = R i_a + L \frac{di_a}{dt} + e_a \quad (1)$$

$$V_b = R i_b + L \frac{di_b}{dt} + e_b \quad (2)$$

$$V_c = R i_c + L \frac{di_c}{dt} + e_c \quad (3)$$

Where $R_a = R_b = R_c = R$ Stator resistance per Phase [Ohm]

$L_a = L_b = L_c = L$ Stator inductance per Phase [Henry]

$V_a, V_b,$ and V_c are the stator phase voltages [volt]

$i_a, i_b,$ and i_c are stator phase currents. [Ampere]

$e_a, e_b,$ and e_c are motor back EMFs and the expression of $e_a, e_b,$ and e_c will be given in equations 5,6 and 7.

The modeling of equations of BLDC motor can be represented in matrix form

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} R + pL & 0 & 0 \\ 0 & R + pL & 0 \\ 0 & 0 & R + pL \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (4)$$

Where p is the matrix presents $\frac{d}{dt}$

When BLDC motor rotates each winding generates a voltage known as Back EMF which opposes the main voltage supplied to the winding according to Lenz's Law. The polarity of the back emf is opposite direction of the source voltage.

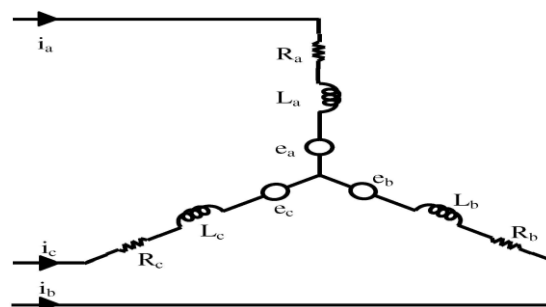


Figure 1. Equivalent circuit model of BLDC motor

$$e_a = k \omega f(\theta) \quad (5)$$

$$e_b = k \omega f(\theta - \frac{2\pi}{3}) \quad (6)$$

$$e_c = k \omega f(\theta + \frac{2\pi}{3}) \quad (7)$$

Where k is the Back EMF constant [v/rad/s].

Θ is the electrical rotor angle

ω is the mechanical speed of the rotor [rad/s]

First, we consider equation no. (1):

$$V_a = R i_a + L \frac{di_a}{dt} + e_a$$

Considering Laplace transform of the above equation we find the transfer function

$$V_a(s) - e_a(s) = R I_a(s) + L s I_a(s) \quad (7)$$

$$\frac{I_a(s)}{V_a(s) - e_a(s)} = \frac{1}{R + Ls} \quad (8)$$

The Laplace transform of equation (2) and (3) are calculated and it is shown in the form of equation (9) and (10):

$$\frac{I_b(s)}{V_b(s) - e_b(s)} = \frac{1}{R + Ls} \quad (9)$$

$$\frac{I_c(s)}{V_c(s) - e_c(s)} = \frac{1}{R + Ls} \quad (10)$$

Here we have considered four pole configurations. The electrical rotor angle is equal to the mechanical rotor angle multiplied by the number of pole pairs:

$$\theta = \frac{p}{2} \theta_m \text{ where } \theta_m \text{ is the mechanical rotor angle.}$$

$$\frac{d\theta}{dt} = \frac{p}{2} \cdot \frac{d\theta_m}{dt} = \frac{p}{2} \cdot \omega$$

$$T_e = \frac{i_a e_a + i_b e_b + i_c e_c}{\omega}$$

$$= i_a \cdot k \cdot f(\theta) + i_b \cdot k \cdot f(\theta - \frac{2\pi}{3}) + i_c \cdot k \cdot f(\theta + \frac{2\pi}{3}) \quad (12)$$

$$T_e = J \frac{d\omega}{dt} + T_L + B \omega \quad (13)$$

$$T_e - T_L = J \frac{d\omega}{dt} + B \omega$$

$$\frac{\omega(s)}{T_e(s) - T_L(s)} = \frac{1}{Js + \theta} \quad (14)$$

Schematic Block Diagram of Motor

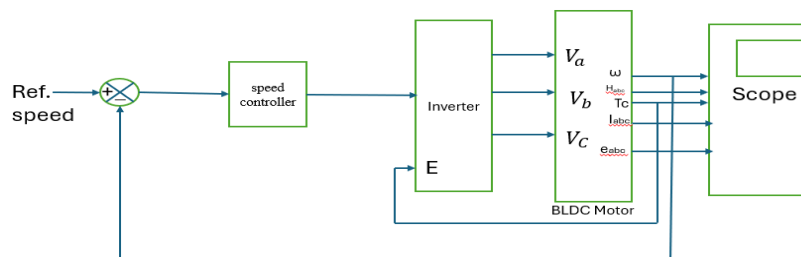


Figure 2. Schematic Block Diagram of BLDC Motor

Advantages of Mathematical Modelling for BLDC Speed Control:

Precise System Understanding

A mathematical model captures the dynamic behavior of the motor (electrical and mechanical subsystems), allowing accurate prediction of performance.

Design of Advanced Controllers

Enables the use of control techniques like PID, Model Predictive Control (MPC), or Field-Oriented Control (FOC) that require a model to function effectively.

Simulation and Testing

Before actual implementation, the model allows testing of various control strategies using simulation (e.g., MATLAB/Simulink), reducing hardware testing time and risk.

Optimization

Helps in tuning control parameters (gains, bandwidth, etc.) to achieve desired performance like fast response, minimal overshoot, or energy efficiency. Fault Detection and Diagnosis

A model-based approach allows for comparison of expected and actual behaviour, helping to identify faults like rotor misalignment or sensor failure.

Scalability and Adaptability

Once a mathematical model is created, it can be scaled or adapted for motors of different ratings or integrated into more complex systems like robotics or EVs.

Supports Sensor less Control

Many sensor less techniques (like back-EMF estimation) rely on an accurate motor model to estimate rotor position and speed.

Conclusion

The paper provides a new idea to control the speed of BLDC Motor with accuracy; also, this model reduces torque ripples, so the efficiency is improved.

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